

EXP4

April 10, 2026

```
[2]: import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
from sklearn.pipeline import Pipeline
from sklearn.metrics import adjusted_rand_score
from sklearn.preprocessing import StandardScaler
```

```
[4]: iris = load_iris()
X = iris.data
y = iris.target

print(X.shape)
```

(150, 4)

```
[6]: scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```
[8]: kmeans_raw = KMeans(n_clusters=3, random_state=42)
labels_raw = kmeans_raw.fit_predict(X_scaled)

ari_raw = adjusted_rand_score(y, labels_raw)
print("ARI without PCA:", ari_raw)
```

ARI without PCA: 0.432804702527474

```
[10]: pipeline = Pipeline([
    ('pca', PCA(n_components=2)),
    ('kmeans', KMeans(n_clusters=3, random_state=42))
])

labels_pca = pipeline.fit_predict(X_scaled)

ari_pca = adjusted_rand_score(y, labels_pca)
print("ARI with PCA:", ari_pca)
```

ARI with PCA: 0.432804702527474

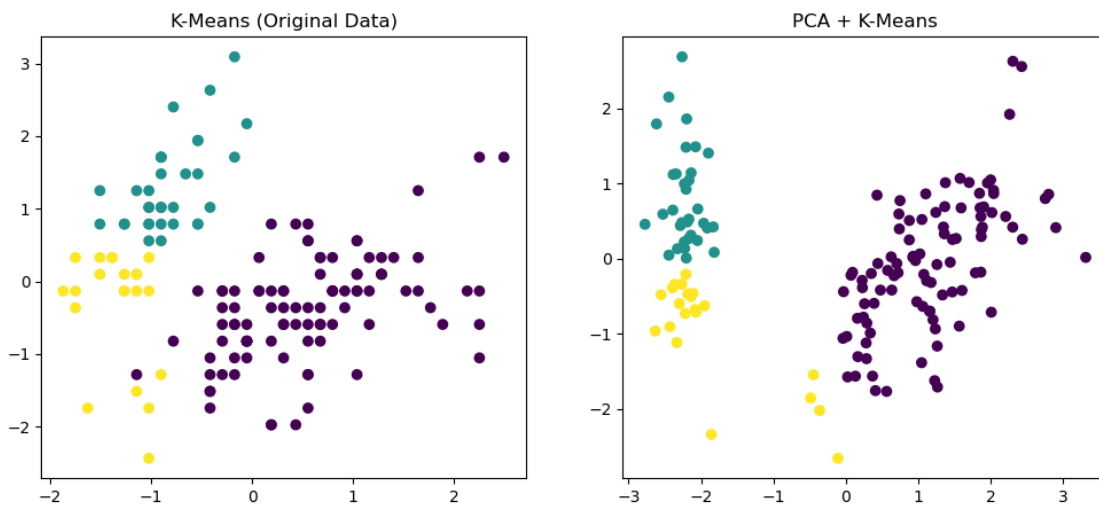
```
[12]: # Transform data for plotting
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

plt.figure(figsize=(12,5))

# Before PCA (using first 2 features for visualization)
plt.subplot(1,2,1)
plt.scatter(X_scaled[:,0], X_scaled[:,1], c=labels_raw)
plt.title("K-Means (Original Data)")

# After PCA
plt.subplot(1,2,2)
plt.scatter(X_pca[:,0], X_pca[:,1], c=labels_pca)
plt.title("PCA + K-Means")

plt.show()
```



```
[ ]:
```